Credal Variable Elimination: Exact Marginal Inference for Credal Networks

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Abstract

Credal Variable Elimination (CVE) is an exact algorithm for computing lower and upper bounds on marginal probabilities in credal networks. It extends the classical variable elimination algorithm for Bayesian networks to operate on *potentials*—finite sets of real-valued functions representing the extreme points of local credal sets. By combining, marginalizing, and pruning these potentials along an elimination ordering, CVE produces tight probability intervals for any query variable given evidence.

1 Preliminaries

1.1 Credal Networks

A **credal network** is a pair $\langle G, \mathcal{K} \rangle$ where G is a directed acyclic graph (DAG) over variables $\mathbf{X} = \{X_1, \ldots, X_n\}$ with finite domains $\mathcal{D} = \{D_1, \ldots, D_n\}$, and $\mathcal{K}$ is a collection of conditional credal sets $K(X_i \mid \mathrm{Pa}(X_i) = \pi)$, one for each variable X_i and each configuration π of its parents $\mathrm{Pa}(X_i)$ in G.

Each credal set $K(X_i \mid \pi)$ is a closed convex set of probability distributions over X_i, specified by its *extreme points* (vertices):

$$\mathrm{ext}\, K(X_i \mid \pi) = \{P_1(X_i \mid \pi), \ldots, P_v(X_i \mid \pi)\}.$$

1.2 Strong Extension

The **strong extension** of a separately specified credal network is the convex hull of all joint distributions formed by combining one extreme point per local credal set:

$$\mathrm{co}\, K(\mathbf{X}) = \mathrm{CH}\Big\{ P(\mathbf{x}) = \prod_{i=1}^{n} P(x_i \mid \pi_i) : P(X_i \mid \pi_i) \in \mathrm{ext}\, K(X_i \mid \pi_i) \Big\}. \tag{1}$$

Theorem 1 (Mauá & Cozman, 2020)**.** *Any combination of extreme points from the local credal sets is an extreme point of the strong extension. Thus the strong extension can be built directly from the local extreme points.*

1.3 Marginal Inference

Given a query variable Z and evidence $\mathbf{Y} = \mathbf{y}$, the goal is to compute:

$$\underline{P}(Z = z \mid \mathbf{Y} = \mathbf{y}) = \min_{P \in \mathrm{ext}\, K(\mathbf{X})} \frac{\sum_{\mathbf{x}'} \prod_i P(x_i \mid \pi_i)}{\sum_{z'} \sum_{\mathbf{x}'} \prod_i P(x_i \mid \pi_i)}, \tag{2}$$

where the sums in numerator and denominator are over the marginalized variables $\mathbf{X}' = \mathbf{X} \setminus (\{Z\} \cup \mathbf{Y})$, and similarly for the upper probability $\overline{P}$ with max.

2 Potentials

Definition 1 (Potential)**.** *A **potential** $\phi(\mathbf{Y})$ over variables $\mathbf{Y}$ is a finite set of non-negative real-valued functions $\{f_1, f_2, \ldots, f_m\}$ on the joint domain of $\mathbf{Y}$. Each function f_k maps configurations of $\mathbf{Y}$ to $\mathbb{R}_{\geq 0}$.*

Three operations on potentials are central to the algorithm:

Product. The product of $\phi(\mathbf{Y})$ and $\psi(\mathbf{Z})$ is:

$$\phi \cdot \psi = \{f \cdot g : f \in \phi,\ g \in \psi\},$$

where $(f \cdot g)(\mathbf{y}, \mathbf{z}) = f(\mathbf{y}) \cdot g(\mathbf{z})$ over the joint domain $\mathbf{Y} \cup \mathbf{Z}$ (with broadcasting for non-shared variables).

Sum-Marginal. The sum-marginal of $\phi(\mathbf{Y})$ with respect to $X \subseteq \mathbf{Y}$ is:

$$\sum_X \phi(\mathbf{Y}) = \Big\{\sum_x f(\mathbf{y}) : f \in \phi\Big\},$$

producing functions over $\mathbf{Y} \setminus \{X\}$.

Pruning. The pruning operation removes dominated functions:

$$\text{prune}(\phi) = \{f \in \phi : \nexists\, g \in \phi \setminus \{f\} \text{ s.t. } g(\mathbf{y}) \geq f(\mathbf{y})\ \forall\, \mathbf{y}\}.$$

This reduces the cardinality of potentials without affecting the min/max bounds.

3 Algorithm

Algorithm 1 Credal Variable Elimination (CVE)

Require: Extreme points $\text{ext}\, K(X_i \mid \text{Pa}(X_i))$ for each node X_i, query variable Z, evidence $\mathbf{Y} = \mathbf{y}$
Ensure: Lower and upper bounds $[\underline{P}(Z{=}z \mid \mathbf{y}),\ \overline{P}(Z{=}z \mid \mathbf{y})]$ for each state z

$\Gamma \leftarrow \emptyset$ $\triangleright$ Set of potentials
for each variable $X_i \in \mathbf{X}$ **do**
  $\phi_{X_i} \leftarrow \{p : p \in \text{ext}\, K(X_i \mid \text{Pa}(X_i))\}$
  $\Gamma \leftarrow \Gamma \cup \{\phi_{X_i}\}$
end for
for each evidence variable $Y_j \in \mathbf{Y}$ **do**
  $\phi'_{Y_j} \leftarrow \{\delta_{y_j}\}$ $\triangleright$ Indicator function
  $\Gamma \leftarrow \Gamma \cup \{\phi'_{Y_j}\}$
end for

Compute elimination ordering $\mathbf{o} = (X_{\sigma(1)}, \ldots, X_{\sigma(n-1)})$ excluding Z
for each X_i in ordering $\mathbf{o}$ **do**
  $\Gamma_{X_i} \leftarrow \{\phi \in \Gamma : X_i \in \text{scope}(\phi)\}$
  $\Gamma \leftarrow \Gamma \setminus \Gamma_{X_i}$
  $\psi \leftarrow \prod\{\phi \in \Gamma_{X_i}\}$ $\triangleright$ Combine all potentials in bucket
  $\psi' \leftarrow \sum_{X_i} \psi$ $\triangleright$ Marginalize out X_i
  $\psi'' \leftarrow \text{prune}(\psi')$ $\triangleright$ Remove dominated functions
  $\Gamma \leftarrow \Gamma \cup \{\psi''\}$
end for

$\phi_{\text{final}} \leftarrow \prod\{\phi \in \Gamma\}$ $\triangleright$ Remaining potentials over Z
$\underline{P}(z) \leftarrow 1,\ \overline{P}(z) \leftarrow 0$ for each state z
for each $f \in \phi_{\text{final}}$ **do**
  $\mathbf{p} \leftarrow f / \sum_{z'} f(z')$ $\triangleright$ Normalize
  for each state z **do**
    $\underline{P}(z) \leftarrow \min(\underline{P}(z),\ \mathbf{p}(z))$
    $\overline{P}(z) \leftarrow \max(\overline{P}(z),\ \mathbf{p}(z))$
  end for
end for
return $[\underline{P}(z),\ \overline{P}(z)]$ for each z

3.1 Elimination Ordering

Two heuristics are supported:

- **Topological order**: Variables are eliminated in topological order of the DAG (non-evidence, non-query variables first, then evidence variables).
- **Min-fill**: At each step, eliminate the variable that introduces the fewest new edges (fill-in) in the interaction graph. This heuristic minimizes the width of the resulting tree decomposition.

4 Running Example

Example 1 (Alarm Network)**.** *Consider the LCN with variables B (Burglary), E (Earthquake), A (Alarm), and compound variable CD (representing C and D). The DAG structure is:*

$$B \to A, \quad E \to A, \quad A \to CD.$$

The local credal sets (extreme points from LRS vertex enumeration) are:

Node	***Parent Config***	***Vertices***
B	—	$[0.8, 0.2]$, $[0.9, 0.1]$
E	—	$[0.9, 0.1]$, $[0.95, 0.05]$
A	$B{=}0, E{=}0$	$[0.956, 0.044]$, $[1.0, 0.0]$
A	$B{=}1, E{=}0$	$[0.0, 1.0]$, $[1.0, 0.0]$
A	$B{=}0, E{=}1$	$[0.0, 1.0]$, $[1.0, 0.0]$
A	$B{=}1, E{=}1$	$[0.0, 1.0]$, $[1.0, 0.0]$

Query: $P(B{=}1)$ **(no evidence).** *Elimination order (topological): E, A, CD.*

***Step 1: Eliminate** E. Potentials involving E: ϕ_E (scope $\{E\}$, 2 functions) and ϕ_A (scope $\{A, B, E\}$, $2^3 = 8$ functions for the separately specified A). Combine: $\psi = \phi_E \cdot \phi_A$ produces $2 \times 8 = 16$ functions over $\{A, B, E\}$. Marginalize E: sum over E's domain, producing 16 functions over $\{A, B\}$. Prune: remove dominated functions.*

***Step 2: Eliminate** A. Combine with CD factor, marginalize A, prune.*

***Step 3: Eliminate** CD. Marginalize CD, leaving functions over $\{B\}$ only.*

***Result.** Each surviving function $f(B)$ is normalized: $p(B{=}1) = f(1)/(f(0) + f(1))$. The minimum across all functions gives $\underline{P}(B{=}1) = 0.10$ and the maximum gives $\overline{P}(B{=}1) = 0.20$.*

5 Complexity

Theorem 2. *The time and space complexity of CVE is $O(n \cdot C \cdot k^{w^*})$, where n is the number of variables, k bounds the domain sizes, w^* is the induced width of the elimination ordering, and C bounds the cardinality of the intermediate potentials.*

In the worst case, C is exponential in the number of extreme points. However, pruning keeps C manageable in practice for networks of moderate treewidth.

6 References

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3. F.G. Cozman. Credal networks. *Artificial Intelligence*, 120(2):199–233, 2000.